6 Выборка данных из базы данных. Выполнение запроса ROLLUP, GROUPING SETS, CUBE.

1. Количество продаж по району, риэлтору и общее

```
SELECT
    d.district_name,
    r.realtor_id,
    COUNT(s.sale_id) AS sales_count
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
JOIN districts d ON re.district_id = d.district_id
JOIN realtor r ON s.realtor_id = r.realtor_id
GROUP BY ROLLUP (d.district_name, r.realtor_id)
ORDER BY d.district_name NULLS LAST, r.realtor_id NULLS LAST;
```

```
transport=# ^C
transport=# SELECT
    d.district_name,
    r.realtor_id,
    COUNT(s.sale_id) AS sales_count
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
JOIN districts d ON re.district_id = d.district_id
JOIN realtor r ON s.realtor_id = r.realtor_id
GROUP BY ROLLUP (d.district_name, r.realtor_id)
ORDER BY d.district_name NULLS LAST, r.realtor_id NULLS LAST;
 district_name | realtor_id | sales_count
---------------+------------+-------------
 Западный      |          3 |           1
 Западный      |          4 |           1
 Западный      |          5 |           1
 Западный      |            |           3
 Северный      |          1 |           1
 Северный      |          2 |           2
 Северный      |          3 |           1
 Северный      |          4 |           1
 Северный      |          5 |           1
 Северный      |            |           6
 Центральный   |          1 |           2
 Центральный   |          2 |           1
 Центральный   |          3 |           1
 Центральный   |          4 |           1
 Центральный   |          5 |           1
 Центральный   |            |           6
 Южный         |          1 |           1
 Южный         |          2 |           1
 Южный         |          3 |           1
 Южный         |          4 |           1
 Южный         |          5 |           1
 Южный         |            |           5
               |            |          20
(23 rows)

transport=#
```

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2. Средняя стоимость 1м² по району и этажу с заменой NULL

```sql
SELECT
    COALESCE(d.district_name, 'Итого по району') AS district,
    COALESCE(CAST(re.floor AS TEXT), 'Итого по этажу') AS floor,
    AVG(s.sale_price / re.area) AS avg_price_per_m2
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
```

```
JOIN districts d ON re.district_id = d.district_id
GROUP BY ROLLUP (d.district_name, re.floor)
ORDER BY d.district_name NULLS LAST, re.floor NULLS LAST;
```

```
transport=# SELECT
    COALESCE(d.district_name, 'Итого по району') AS district,
    COALESCE(CAST(re.floor AS TEXT), 'Итого по этажу') AS floor,
    AVG(s.sale_price / re.area) AS avg_price_per_m2
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
JOIN districts d ON re.district_id = d.district_id
GROUP BY ROLLUP (d.district_name, re.floor)
ORDER BY d.district_name NULLS LAST, re.floor NULLS LAST;
    district     |     floor      |  avg_price_per_m2
-----------------+----------------+--------------------
 Западный        | 2              |             200000
 Западный        | 3              | 170833.33333333334
 Западный        | 5              | 233333.33333333334
 Западный        | Итого по этажу |  201388.8888888889
 Северный        | 2              |  83333.33333333333
 Северный        | 3              |  306666.6666666667
 Северный        | 4              | 234615.38461538462
 Северный        | 6              |  176913.5294117647
 Северный        | 7              |             120000
 Северный        | Итого по этажу | 192690.71644042231
 Центральный     | 3              |  288461.5384615385
 Центральный     | 4              | 414285.71428571426
 Центральный     | 5              | 195888.15789473685
 Центральный     | 10             | 116666.66666666667
 Центральный     | Итого по этажу | 249941.96227748864
 Южный           | 1              |  67857.14285714286
 Южный           | 2              |  92857.14285714286
 Южный           | 6              | 228571.42857142858
 Южный           | 7              |               80000
 Южный           | 8              |             200000
 Южный           | Итого по этажу | 133857.14285714287
 Итого по району | Итого по этажу | 196462.42266299232
(22 rows)

transport=#
```

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3. Суммарная продажная стоимость по типу, району, этажу

```
SELECT
    t.type_name,
    d.district_name,
    re.floor,
    SUM(s.sale_price) AS total_price
FROM sale s
```

```
JOIN real_estate re ON s.estate_id = re.estate_id
JOIN type t ON re.type_id = t.type_id
JOIN districts d ON re.district_id = d.district_id
GROUP BY GROUPING SETS (
    (t.type_name),
    (t.type_name, d.district_name),
    (t.type_name, d.district_name, re.floor),
    ()
)
ORDER BY t.type_name NULLS LAST, d.district_name NULLS LAST, re.floor
NULLS LAST;
```

```
transport=# SELECT
    t.type_name,
    d.district_name,
    re.floor,
    SUM(s.sale_price) AS total_price
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
JOIN type t ON re.type_id = t.type_id
JOIN districts d ON re.district_id = d.district_id
GROUP BY GROUPING SETS (
    (t.type_name),
    (t.type_name, d.district_name),
    (t.type_name, d.district_name, re.floor),
    ()
)
ORDER BY t.type_name NULLS LAST, d.district_name NULLS LAST, re.floor NULLS LAST;
  type_name  | district_name | floor | total_price
-------------+---------------+-------+-------------
 Апартаменты | Западный      |     2 |    22000000
 Апартаменты | Западный      |     3 |    20500000
 Апартаменты | Западный      |       |    42500000
 Апартаменты |               |       |    42500000
 Дом         | Северный      |     2 |    12500000
 Дом         | Северный      |       |    12500000
 Дом         | Южный         |     1 |     9500000
 Дом         | Южный         |     2 |    13000000
 Дом         | Южный         |     7 |    12000000
 Дом         | Южный         |       |    34500000
 Дом         |               |       |    47000000
 Квартира    | Западный      |     5 |    21000000
 Квартира    | Западный      |       |    21000000
 Квартира    | Северный      |     3 |    23000000
 Квартира    | Северный      |     4 |    27500000
 Квартира    | Северный      |     6 |    15037650
 Квартира    | Северный      |     7 |     9000000
 Квартира    | Северный      |       |    74537650
 Квартира    | Центральный   |     3 |    35500000
 Квартира    | Центральный   |     4 |    29000000
 Квартира    | Центральный   |     5 |    34500000
 Квартира    | Центральный   |    10 |    10500000
 Квартира    | Центральный   |       |   109500000
 Квартира    | Южный         |     6 |    16000000
 Квартира    | Южный         |     8 |    13000000
 Квартира    | Южный         |       |    29000000
 Квартира    |               |       |   234037650
             |               |       |   323537650
(28 rows)

transport#
```

4. Максимальная разница между заявленной и проданной стоимостью за текущий год

```sql
SELECT
    d.district_name,
    r.realtor_id,
    MAX(s.sale_price - re.price) AS max_diff
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
JOIN districts d ON re.district_id = d.district_id
JOIN realtor r ON s.realtor_id = r.realtor_id
WHERE EXTRACT(YEAR FROM s.sale_date) = EXTRACT(YEAR FROM CURRENT_DATE)
GROUP BY ROLLUP (d.district_name, r.realtor_id)
HAVING MAX(s.sale_price - re.price) > 0
ORDER BY d.district_name NULLS LAST, r.realtor_id NULLS LAST;
```

```
transport=# SELECT
    d.district_name,
    r.realtor_id,
    MAX(s.sale_price - re.price) AS max_diff
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
JOIN districts d ON re.district_id = d.district_id
JOIN realtor r ON s.realtor_id = r.realtor_id
WHERE EXTRACT(YEAR FROM s.sale_date) = EXTRACT(YEAR FROM CURRENT_DATE)
GROUP BY ROLLUP (d.district_name, r.realtor_id)
HAVING MAX(s.sale_price - re.price) > 0
ORDER BY d.district_name NULLS LAST, r.realtor_id NULLS LAST;
 district_name | realtor_id | max_diff
---------------+------------+----------
 Северный      |          2 |  9000000
 Северный      |            |  9000000
 Центральный   |          3 |  3000000
 Центральный   |            |  3000000
 Южный         |          2 |  3500000
 Южный         |          3 |  3000000
 Южный         |            |  3500000
               |            |  9000000
(8 rows)

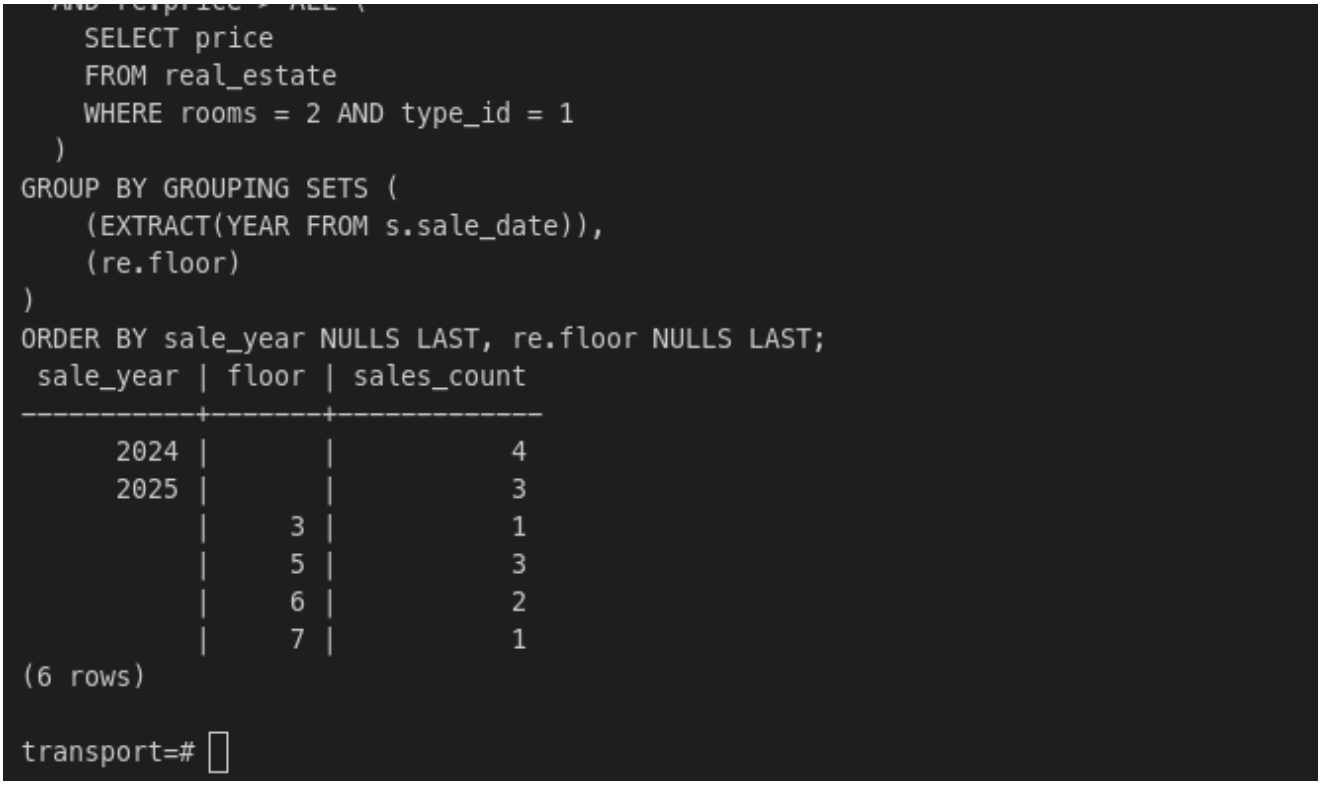
transport=#
```

---

5. Количество продаж двухкомнатных объектов по году и этажу

```sql
SELECT
    EXTRACT(YEAR FROM s.sale_date) AS sale_year,
```

```
    re.floor,
    COUNT(s.sale_id) AS sales_count
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
WHERE re.rooms = 2
  AND re.price > ALL (
    SELECT price
    FROM real_estate
    WHERE rooms = 2 AND type_id = 1
  )
GROUP BY GROUPING SETS (
    (EXTRACT(YEAR FROM s.sale_date)),
    (re.floor)
)
ORDER BY sale_year NULLS LAST, re.floor NULLS LAST;
```



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6. Средняя стоимость 1м² с проверкой площади

```
SELECT
    EXTRACT(YEAR FROM s.sale_date) AS sale_year,
    d.district_name,
    AVG(s.sale_price / re.area) AS avg_price_per_m2
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
JOIN districts d ON re.district_id = d.district_id
WHERE re.area = (
```

```
    SELECT SUM(area)
    FROM real_estate_structure res
    WHERE res.estate_id = re.estate_id
)
GROUP BY GROUPING SETS (
    (EXTRACT(YEAR FROM s.sale_date)),
    (EXTRACT(YEAR FROM s.sale_date), d.district_name),
    ()
)
ORDER BY sale_year NULLS LAST, d.district_name NULLS LAST;
```

```
transport=# SELECT
    EXTRACT(YEAR FROM s.sale_date) AS sale_year,
    d.district_name,
    AVG(s.sale_price / re.area) AS avg_price_per_m2
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
JOIN districts d ON re.district_id = d.district_id
WHERE re.area = (
    SELECT SUM(area)
    FROM real_estate_structure res
    WHERE res.estate_id = re.estate_id
)
GROUP BY GROUPING SETS (
    (EXTRACT(YEAR FROM s.sale_date)),
    (EXTRACT(YEAR FROM s.sale_date), d.district_name),
    ()
)
ORDER BY sale_year NULLS LAST, d.district_name NULLS LAST;
 sale_year | district_name | avg_price_per_m2
-----------+---------------+------------------
      2024 | Центральный   |           290000
      2024 |               |           290000
           |               |           290000
(3 rows)

transport=#
```

---

7. Средняя площадь объектов с фильтром

```
SELECT
    EXTRACT(YEAR FROM s.sale_date) AS sale_year,
    d.district_name,
    ROUND(AVG(re.area), 2) AS avg_area
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
JOIN districts d ON re.district_id = d.district_id
GROUP BY GROUPING SETS (
```

```
    (EXTRACT(YEAR FROM s.sale_date)),
    (d.district_name),
    (EXTRACT(YEAR FROM s.sale_date), d.district_name),
    ()
)
HAVING AVG(re.area) >= 30
ORDER BY sale_year NULLS LAST, d.district_name NULLS LAST;
```

```
transport=# SELECT
    EXTRACT(YEAR FROM s.sale_date) AS sale_year,
    d.district_name,
    ROUND(AVG(re.area), 2) AS avg_area
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
JOIN districts d ON re.district_id = d.district_id
GROUP BY GROUPING SETS (
    (EXTRACT(YEAR FROM s.sale_date)),
    (d.district_name),
    (EXTRACT(YEAR FROM s.sale_date), d.district_name),
    ()
)
HAVING AVG(re.area) >= 30
ORDER BY sale_year NULLS LAST, d.district_name NULLS LAST;
 sale_year | district_name | avg_area
-----------+---------------+----------
      2024 | Западный      |    90.00
      2024 | Северный      |    91.25
      2024 | Центральный   |    73.00
      2024 | Южный         |   140.00
      2024 |               |    87.27
      2025 | Западный      |   115.00
      2025 | Северный      |    70.00
      2025 | Центральный   |    95.00
      2025 | Южный         |   106.25
      2025 |               |    98.89
           | Западный      |   106.67
           | Северный      |    84.17
           | Центральный   |    76.67
           | Южный         |   113.00
           |               |    92.50
(15 rows)

transport=#
```

8. Количество объектов по типу и материалу

```
SELECT
    t.type_name,
    bm.material_name,
```

```
    COUNT(re.estate_id) AS count
FROM real_estate re
JOIN type t ON re.type_id = t.type_id
JOIN building_materials bm ON re.material_id = bm.material_id
GROUP BY CUBE (t.type_name, bm.material_name)
ORDER BY t.type_name NULLS LAST, bm.material_name NULLS LAST;
```

```
transport=# SELECT
    t.type_name,
    bm.material_name,
    COUNT(re.estate_id) AS count
FROM real_estate re
JOIN type t ON re.type_id = t.type_id
JOIN building_materials bm ON re.material_id = bm.material_id
GROUP BY CUBE (t.type_name, bm.material_name)
ORDER BY t.type_name NULLS LAST, bm.material_name NULLS LAST;
  type_name  | material_name | count
-------------+---------------+-------
 Апартаменты | Панель        |     2
 Апартаменты |               |     2
 Дом         | Панель        |     7
 Дом         |               |     7
 Квартира    | Кирпич        |    14
 Квартира    | Панель        |     7
 Квартира    |               |    21
             | Кирпич        |    14
             | Панель        |    16
             |               |    30
(10 rows)

transport=#
```

---

9. Общий объем продаж по районам и типам

```
SELECT
    d.district_name,
    t.type_name,
    SUM(s.sale_price) AS total_sales
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
JOIN districts d ON re.district_id = d.district_id
JOIN type t ON re.type_id = t.type_id
GROUP BY CUBE (d.district_name, t.type_name)
ORDER BY d.district_name NULLS LAST, t.type_name NULLS LAST;
```

```
transport=# SELECT
    d.district_name,
    t.type_name,
    SUM(s.sale_price) AS total_sales
FROM sale s
JOIN real_estate re ON s.estate_id = re.estate_id
JOIN districts d ON re.district_id = d.district_id
JOIN type t ON re.type_id = t.type_id
GROUP BY CUBE (d.district_name, t.type_name)
ORDER BY d.district_name NULLS LAST, t.type_name NULLS LAST;
 district_name |  type_name  | total_sales
---------------+-------------+-------------
 Западный      | Апартаменты |    42500000
 Западный      | Квартира    |    21000000
 Западный      |             |    63500000
 Северный      | Дом         |    12500000
 Северный      | Квартира    |    74537650
 Северный      |             |    87037650
 Центральный   | Квартира    |   109500000
 Центральный   |             |   109500000
 Южный         | Дом         |    34500000
 Южный         | Квартира    |    29000000
 Южный         |             |    63500000
               | Апартаменты |    42500000
               | Дом         |    47000000
               | Квартира    |   234037650
               |             |   323537650
(15 rows)

transport=#
```

---

10. Объем продаж по месяцам и годам

```sql
SELECT
    EXTRACT(YEAR FROM s.sale_date) AS sale_year,
    EXTRACT(MONTH FROM s.sale_date) AS sale_month,
    SUM(s.sale_price) AS total_sales
FROM sale s
GROUP BY CUBE (sale_year, sale_month)
HAVING SUM(s.sale_price) > 1000000
ORDER BY sale_year NULLS LAST, sale_month NULLS LAST;
```

```
transport=# SELECT
    EXTRACT(YEAR FROM s.sale_date) AS sale_year,
    EXTRACT(MONTH FROM s.sale_date) AS sale_month,
    SUM(s.sale_price) AS total_sales
FROM sale s
GROUP BY CUBE (sale_year, sale_month)
HAVING SUM(s.sale_price) > 1000000
ORDER BY sale_year NULLS LAST, sale_month NULLS LAST;
 sale_year | sale_month | total_sales
-----------+------------+-------------
      2024 |          2 |    14500000
      2024 |          3 |    11500000
      2024 |          4 |    15037650
      2024 |          5 |    24000000
      2024 |          6 |    13000000
      2024 |          7 |    10500000
      2024 |          8 |    12500000
      2024 |          9 |    21000000
      2024 |         10 |     9000000
      2024 |         11 |    29000000
      2024 |         12 |    16500000
      2024 |            |   176537650
      2025 |          1 |    23000000
      2025 |          2 |    16000000
      2025 |          3 |   108000000
      2025 |            |   147000000
           |          1 |    23000000
           |          2 |    30500000
           |          3 |   119500000
           |          4 |    15037650
           |          5 |    24000000
           |          6 |    13000000
           |          7 |    10500000
           |          8 |    12500000
           |          9 |    21000000
           |         10 |     9000000
           |         11 |    29000000
           |         12 |    16500000
           |            |   323537650
(29 rows)

transport=#
```